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What if India turned vegetarian by 2050?

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I affirm that I have identified all my sources and that no part of my dissertation paper uses unacknowledged materials.

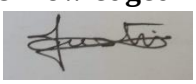


TABLE OF CONTENTS

Abstract……i
Keywords……i
JEL Classification……i

Chapter 1: INTRODUCTION……..1

Chapter 2: LITERATURE REVIEW……...2

2.1 History of the ‘’diet problem’’……..…2

2.2 Research on the need, possibility and benefits of switching to a plant-based diet.3

2.2.1 Stehfest et al.[11] (2009)……3

2.2.2 Godfray et al.[8] (2010)……..4

2.2.3 Popp, Lotze-Campen and Bodirsky [6] (2010)……...5

2.2.4 Bajlezj et al.[7] (2014)……6

2.2.5 Hallstrom, Carlsson-Kanyama and Borjesson [9](2014)……7

2.2.6 Tilman and Clark[4] (2014)……8

2.2.7 Hedenus, Wirsenius and Johansson[5] (2014)……9

2.2.8 Springmann et al.[17] (2016)……...9

Chapter 3: THE POSSIBILITY OF INDIA TURNING VEGAN BY 2050……9

Chapter 4: METHODS AND DATA……..10

4.1 Overview of data and methodology……...10

4.2 Detailed Methodology……12

Chapter 5: RESULTS AND DISCUSSION……12

Chapter 6: CONCLUSION……..13

Chapter 7: APPENDIX……14

7.1 Diet under consideration……14

7.2 Energy Requirements and Price data……..14

7.3 Linear Programming Analysis……16

Chapter 8: REFERENCES……..17

Abstract

In developed countries such as the United States of America and the United Kingdom, the number of people choosing plant-based diets has increased dramatically in recent years. This appears to indicate a greater understanding of the linkages between meat intake and diseases, as well as the environmental consequences of meat consumption. With every passing year inflicting devastation on the environment, a plant-based diet may become the norm in the future.

The year 2050 is distant enough in the future in this study for such a scenario to be true in India. Using linear programming methods, this research investigates the feasibility of a completely vegan diet for the ordinary Indian. According to the findings, a dietary pattern like this may be out of reach for people earning less than 1.9 percent of the national average income in 2050, compared to 0.08 percent in 2020. This research also looks at some of the other consequences of the country's citizens' near-complete vegetarianism.

Keywords: Linear Programming, Mathematical Programming, Diet, Health, Nutrition, Food Price, Greenhouse Gas

JEL Classification: C610, I120, Q110, Q540

1. Introduction

In traditional modelling, especially in economics, the impact of or from only a few variables is investigated, with the ceteris paribus criterion used to exclude factors not under investigation. In contrast, scientists at the Massachusetts Institute of Technology constructed World3, a computer that employed system dynamics to encompass nearly all potential economic variables in an attempt "to examine how the world's problems may be simulated on a computer." (1)

Because of several critical variables, one of which is pollution, the study came to the conclusion that society as we know it will not exist by the year 2040. Our food habits account for about 19-29 percent of global greenhouse gas emissions (GHG), with cattle and agriculture accounting for up to 80 percent (2). In India, total enteric emissions from livestock are estimated to be between 6.17 and 10.4 Tg per year (3).

Several research (4-11) have found that making specific dietary modifications, including reducing or eliminating animal-derived items, has health benefits. This adjustment would reduce global GHG emissions while also easing strain on agricultural land demand. In comparison to technical choices, these improvements may be even more successful in minimising the harmful effects of climate change.

Introduction, literature review, the prospect of India becoming vegan by 2050, methodology and data, results and discussion, conclusion, appendices, and references are the eight chapters of this work.

Chapter 1 is the introduction, which provides the reasoning behind the study. Chapter 2 reviews existing literature on the linear programming methodology, which is to be used for the analysis, along with that of the importance of the need for a dietary change in order for food supplies to be able to meet food demand along with other benefits. Chapter 3 focuses on explaining the rationale behind the possibility of the entire nation

turning to plant-based food by the year 2050. Chapter 4 explains the methodology used to compare the share of population that can afford such a diet between the years 2020 and 2050, the results of which are then discussed in the following chapter. Finally, chapter 8 summarizes the results and identifies areas for further research.

2. Literature Review

The following review is divided into two sections: a brief history of the ‘‘diet problem’’ and research on the need, possibility and benefits of switching to dietary patterns which require less of meat and other animal products

2.1 **History of the ‘’diet problem’’**

The first study aiming to develop acceptable diets using linear programming methods was published in 1959, and the history of the diet problem can be traced back to that year. However, thanks to the efforts of US statistician Jerome Cornfield, the endeavour to find cost-effective diets to suit the nutritional demands of US Army personnel began much earlier during World War II (13).

George J. Stigler is responsible for the "Stigler diet problem," or the diet problem as it is now known, in which he employed optimization techniques to reduce dietary costs given daily allowances of calories, protein, calcium, iron, vitamins A,B1,B2,,C, and Niacin (14). Both the United States and the Soviet Union pursued this topic during WWII. George Dantzig, who devised the simplex method, which provided an accurate mathematical solution to Stigler's diet problem, was a major person in this field (15).

Despite the fact that his approach significantly reduced the problem's complexity, finding the answer took weeks due to the limited processing power available at the time, with only mechanical calculators to rely on. "In practise, diet modelling became practical only with the arrival of computers," Buttriss et al. wrote. Standard spreadsheets now include optimization modules based on linear programming." (16)

2.2 **Research on the need, possibility and benefits of switching to a plant-based diet**

2.2.1 **Stehfest et al.[11] (2009)**

Conventional climate change mitigation programmes frequently overlook the consequences of the livestock sector, which accounts for roughly one-fifth of all GHG emissions and nearly four-fifth of all land accounted for by anthropogenic emissions. This is one of the first studies to look into the feasibility of dietary changes as a way to combat ongoing climate change.

The report examines the worldwide impact of a change in food choices using IMAGE (Integrated Model to Assess the Environment) modelling. For the purposes of the study, four diets were adopted, each with varying degrees of meat and animal product replacement for plant-based alternatives.

The study's most noteworthy finding is that substituting meat and animal products reduces greenhouse gas emissions, especially when combined with other traditional emission reduction measures like afforestation and replanting. It has also been demonstrated that meat and animal products do not need to be completely cut to reap these benefits. Partial substitution can also provide a boost.

Another important consequence is that such substitutes can lower land-use demand, allowing for even more reductions in greenhouse gas emissions. This is only conceivable if the land that becomes available as a result of the substitution is allowed to grow flora that absorbs CO2 from the atmosphere.

2.2.2 **Godfray et al.[8] (2010)**

This research examines a multifaceted strategy for meeting global food demands, which are predicted to reach astronomically high levels by 2050 as a result of rapid population expansion and consumption growth. It views future food production as a

three-pronged challenge: delivering enough food to meet rising demand while employing socially and environmentally sustainable ways, and assuring food for the world's poorest people. Godfray et al. are especially interested in technologies that can allow for long-term manufacturing, such as the ones listed below:

Reducing the yield gap – The difference between realised and actual productivity of land employing currently available technology, management, and genetic material is referred to as the yield gap. There are a variety of causes for the occurrence of a large yield gap, particularly in agriculture, including technical limits and economic factors. It's challenging to come up with ways to close all of these gaps. Instead, such solutions must be tailored to the specific location in question. These come with their own set of difficulties, the most significant of which being a rise in GHG emissions. As a result, they must be properly designed and implemented in order to achieve the desired results.

Increasing the upper limits to production – Using genetic techniques to change crops, which can then acquire a desired trait, is one technique that can be used to achieve this. Drought tolerance, enhanced nitrogen-use efficiency, and photosynthetic efficiency could all be long-term benefits of such modifications.

Reducing food waste- Methods that improve awareness of current technology and practises to mitigate and deal with waste, such as those that result in an increase in awareness of existing technologies and practises to mitigate and deal with waste, are crucial to reducing food waste. Rising food prices, as well as suitable laws, can surely help in this respect.

Changing to a better diet- Reducing consumption and dietary items results in individual health advantages as well as a reduction in greenhouse gas emissions from animals.

Increased aquaculture has also been suggested as a way to boost long-term production. However, because of the release of organic effluents into water bodies, it might harm the environment by becoming a source of genetic contamination or diseases for wild species.

As a result, there is no single approach or strategy for achieving long-term food production that can meet rising demand. It is entirely dependent on the time and location, and can be accomplished with proper organisation and execution.

2.2.3 **Popp, Lotze-Campen and Bodirsky** [(6)] (2010)

This article analyses elements that contribute to non-CO2 GHG emissions, mainly from the agriculture sector, in order to assess anticipated levels of non-CO2 GHG emissions. The MAgPIE model (Model of Agricultural Production and its Impact on the Environment) was used to determine anticipated amounts of these emissions [for additional information on the model, see Lotze-Campen et al., 2008]. For the objectives of this study, the model has been updated so that all spatially explicit agricultural activities are linked to non-CO2 agricultural GHG emissions.

On a global scale, the model projects a 63 percent increase in agricultural non-carbon dioxide emissions by 2055 compared to 1995 data. When dietary modifications are taken into account, however, the outcomes vary dramatically. In 2055, compared to 1995, an increase in meat consumption will result in a 76 percent increase in emissions, whereas a 25 percent decrease in meat consumption will result in a 51 percent reduction in emissions. Other augmented scenarios include technical mitigation combined with an increase in meat consumption, which results in a 13 percent increase in emissions in 2055, and technical mitigation combined with a decrease in meat consumption, which results in the least amount of non-CO2 emissions in 2055 at 2519 CO2 equivalent.

Despite the fact that technological mitigation has a significant ability to lower non-CO2 emissions, adequate dietary adjustments make a larger contribution, with a combination of the two being the most successful.

2.2.4 **Bajlezj et al.**[7] (2014)

This publication is a watershed moment in the scholarly literature on the future demand and supply of agricultural food commodities. It illustrates that, due to the extraordinarily high population at the time, growth of agricultural land and intensification of land use may still leave us with unmet food demand in 2050. It investigates a novel approach to this problem: demand-side mitigation. The paper is divided into two sections:

(i) The first section examines land use changes and agricultural environmental effects, with 2009 as the reference year for the data. A study of agricultural biomass flow has also been conducted, which is necessary in order to investigate demand-side solutions. This research shows that a lopsided increase in demand for inefficient food supply flows will result in the entire system being larger, but at the expense of efficiency.

(ii) The second section forecasts future possibilities as justification for demand-side interventions. Such scenarios essentially demonstrate that, in the absence of demand-side adjustments, supply-side changes may not be sufficient to meet food demands in 2050. Reduced food and agricultural wastes, as well as transitioning to better diets, are suggested demand-side initiatives, which help to reduce the need for agricultural development and offset GHG emissions.

As a result, Bajlezj et al. emphasise the importance of demand-side mitigation as a means of addressing future food shortages—an area that has received little attention in academic literature but that I anticipate will be a hot topic in the near future.

2.2.5 **Hallstrom, Carlsson-Kanyama and Borjesson** [9](2014)

This report examines 14 peer-reviewed studies in order to estimate the environmental impact of dietary changes, which are required in tandem with agricultural advancements to meet rising food demand. In order to objectively comprehend the influence of dietary changes, the trio examines the quantity of greenhouse gas emissions and land use demand for 49 distinct dietary scenarios.

The study presents several important results, which fall into two categories:

(i) Reduction in GHG emissions from dietary changes:

Ruminant meat is identified as a major contributor in the study, and its elimination could result in the greatest decrease in greenhouse gas emissions. Overall, reducing meat and other animal products reduces emissions the most, with the greatest potential for reduction in affluent diets, which are diets with high levels of saturated fat, calories, and cholesterol. However, replacing meat with alternatives that have a large environmental impact, such as cheese, is never the best option. Reduced GHG emissions can also be aided by eating only adequate portions of healthy meals.

(ii) Reduction in demand for land-use from a dietary change:

This, too, is greatly aided by a decrease in meat and other animal products. Cutting down on ruminant meat production can help to reduce the demand for agricultural land once more. In order to lower the need for agricultural land, the study emphasises the importance of improving production techniques and modifying consumption patterns. This study combines analyses from many studies that, when examined separately, would not meet the requirements for inclusion in this review. When taken collectively, however, they provide an overview of the environmental benefits of a dietary change, particularly for a vegan or vegetarian diet.

2.2.6 **Tilman and Clark**[4] (2014)

This research seeks to discover linkages between diets, environmental sustainability, and health based on eating trends using data from 100 of the world's most populated nations. It then uses the trends in the data to forecast future nutritional and environmental scenarios.

The study identifies three key themes in dietary changes as income rises:

(i) With the exception of India, an increase in income is accompanied by a dramatic increase in demand for meat proteins such as those from ruminants and pork.

(ii) As income rises, the demand for calories obtained from alcohol, oils, fats, and refined sweets rises as well.

(iii) Increases in income are associated with an increase in per capita calorie demand. However, because part of this food is thrown away, precise consumption numbers are impossible to determine.

According to the findings, current food patterns may raise the risk of non-communicable diseases like coronary heart disease and type-2 diabetes in the future. Switching to popular diets like the Mediterranean or pescaterian diet, on the other hand, can improve health outcomes. Overall, credible diet solutions are required to address the trinity of nutrition, health, and the environment at the same time.

2.2.7 **Hedenus, Wirsenius and Johansson**[5] (2014)

Increases in livestock productivity, technical advancements, and changes in food habits are all considered in this research as approaches to reduce GHG emissions from agriculture. It also uses the 2°C threshold to see if these emission levels are feasible. The analysis is carried out for the years 2000, 2030, 2050, and 2070, with a focus on the last two.

The GHG emissions estimates for 2070 are found to be significantly higher than the

CO2-equivalent emissions for the 2°C objective. Emissions can be brought closer to the objective by proper implementations resulting from increased cattle productivity and the use of technological innovations that can reduce emissions. When a reduction in meat and food consumption is included in, the scenario shifts radically, with a far better chance of attaining the target.

2.2.8 **Springmann et al.**[17] (2016)

Springmann et al., in one of the most important studies to date, looked at the health and environmental impacts of switching to a diet that was lower in animal products. The economic worth of these improvements is also estimated in this study assuming these changes occur by 2050.

3. The prospect of India turning vegan by 2050

Over the last few years, developed countries, particularly the United States of America and the United Kingdom, have seen a rapid increase in plant-based dietary options. Vegans in the United Kingdom have increased from around 150,000 in 2014 to 600,000 in 2019. (18). In the United States, a research by Ipsos Retail Performance found a 300 percent increase in the number of consumers choosing plant-based diets from 2004 to 2009. (19).

Although there are several factors contributing to this, such as increased awareness of the health and environmental effects of meat consumption, the introduction of PBMAs (Plant-Based Meat Alternatives) into the market, which are designed to be near-perfect substitutes for meat products in terms of appearance, taste, and texture, has been a significant contributor (20).

Several studies have found that most PBMA categories are healthier than their meat counterparts, however they have been chastised for their high salt content, which

exceeds most salt target levels (20, 21). PBMAs are still in their infancy in India, despite being a major participant in food markets in the United States and the United Kingdom. However, it is predicted to expand at a CAGR of about 7.48 percent from 2021 to 2026, reaching USD 45.57 million (22).

With the world's biggest number of vegetarians (almost 38 percent of the population), India could have a near-completely vegan future, especially with the introduction of PBMAs (23).

4. Methods and data

4.1 **Overview of data and methodology**

The analysis is conducted for an individual weighing 60 kg (the average weight of an Indian male and female in 2020 is 65kg and 55kg respectively) and aged 18-29.9 years, having a PAL factor of 1.45, associated with moderate levels of physical activity (24).

The diet under consideration is entirely vegan, as opposed to vegetarian, because the latter allows for the eating of animal products without requiring any kind of killing or slaughter, whilst the former prohibits the intake of any meat or other animal products. The likelihood of PBMAs nearly completely substituting for meat supports this hypothesis. This diet is based on WHO/FAO Expert Consultations on Nutrition, Human Energy (26) and Diet (25) criteria, as well as the World Cancer Research Fund's global benchmarks for eating healthy (27). Based on a blog post by Harvard Health Publishing Staff at Harvard Health, an extra demand for protein has been added (28).

The lowest level of per capita income required to buy such a diet has been established based on projected costs of 14 food categories for 2050. This is then contrasted to the present minimal per capita income required to afford a diet like this. The list of food

items under review, as well as their nutritional information, is presented in Table 1.

Nutritional Content(per 100g)	Protein(in grams)	Fat(in grams)	Carbohydrates(in grams)	Fibre(in grams)
Rice[29]	7.3	2.2	71.1	6.8
Wheat[29]	10.6	1.9	61.6	18.9
Potato[29]	2	0.1	15.4	4.4
Lentils[29]	9.02	0.38	20.1	7.9
Milk[30]	3.6	3.8	4.6	
Groundnut oil[29]	_	93.4	_	_
Mustard oil[29]	_	100	_	_
Soybean oil[29]	_	94.6	_	_
Sunflower oil[29]	_	93.2	_	_
Jaggery[29]	0.4	0.1	_	_
Iodized Salt[29]	Sodium=38700mg			
Onion[29]	0.94	0.1	9.93	3.97
Tomato[29]	0.7	0.42	3.84	1
Sugar[29]	_	_	99.6	_

Table 1: Nutritional Content for the 14 food items under consideration

4.2 **Detailed Methodology**

The analysis is carried out in such a way that the person under consideration tries to meet his or her nutritional needs on a weekly basis. In India, CPI inflation rates from 1963 to 2022 (31) were used to calculate the CPI inflation rate for 2050. This is then coupled with 2022 per kg pricing to arrive at 2050 prices for the foods in table 1.

This information is then used to generate a six-constraint optimization problem. This problem was solved using the simplex algorithm, which is used to tackle linear programming problems because of its linearity (refer to Appendix for details).

5. Results and Discussion

According to the calculations, the diet in question is affordable to people earning more than the average per capita GDP in 2020 and 2050. Even at lower levels of per capita income, this holds true. It becomes unaffordable once it is reduced to less than 1.9 percent of the average income in 2050 and 0.08 percent of the average income in 2020. It should be noted that this does not imply that the diet will become too expensive for a bigger proportion of the population in 2050, as such a conclusion can only be drawn based on information of the country's income distribution at the time.

In 2021, the richest 10% of India's population held 57% of the national income, while the poorest 50% had only 13%. (World Inequality Report 2022). Combining this with the findings of our study, we can only conclude that the fraction of the population that can presently afford a vegan diet is substantially higher than in 2050.

This research does not take into consideration the fact that when income rises, so does food spending. Food supplies are expected to suffice in 2050 to meet demand.

6. Conclusion

According to the analysis undertaken as part of this study, at least 1.9 percent of per

capita Income is necessary in 2050, compared to only 0.08 percent in 2020. As a result, consumer inequality is expected to rise during the next 30 years, from 2020 to 2050. This negative result contrasts with numerous research that found positive results in terms of health, the environment, and other economic aspects.

The absence of literature on inequality levels in 2050 is a key restriction in this context. If inequality is set to develop dramatically by 2050 as a result of present food trends, switching to a vegan diet may help to mitigate part of this rise. In addition, further research is needed in the areas of health, the environment, and the economy, all of which will be affected by such a transformation.

7. Appendix

7.1. **Diet under consideration**

The dietary scenario evaluated for this study is mostly based on global guidelines on healthy eating released by "WHO/FAO Expert Consultations on diet, nutrition, and human energy requirements, as well as World Cancer Research Fund recommendations. It is recommended to consume at least five portions (400 g) of fruit and vegetables per day, less than 10% of total energy intake from free sugars, less than 30% of total energy intake from fats, less than 300 g of red meat per week (very little, if any, is processed), and not to exceed recommended total energy intake."(17)

Carbohydrate and protein requirements have also been added to the mix. The Institute of Medicine (IOM) established a carbohydrate intake of 130 grammes per day for adults in a report published in 2002. (32). In addition, a Harvard Health Publishing Staff blog entry establishes a daily protein need of 0.8 grammes per kilogramme of body weight (28).

7.2. **Energy requirements and Price data**

According to the National Institute of Nutrition in India's 2020 report, the average weight of an average Indian male is 65 kilogrammes, and the average weight of an average Indian

female is 55 kilogrammes (24). As a result, we used the arithmetic mean of these two for the sake of our research, focused on a person weighing roughly 60kg. The daily energy requirement for someone with a PAL factor of 1.45 (i.e., someone who lives a moderately active lifestyle) and is between the ages of 18 and 29.9 is 36 kcal per kg of body weight, which equates to a weekly energy requirement of about 15120 kcal, according to FAO recommendations (33).

The prices for the 14 food items under consideration in this study were acquired from price data published for March 11, 2020 by the Department of Consumer Affairs, Ministry of Consumer Affairs, Food & Public Distribution, Government of India (34). Using statistics from the Organisation for Economic Co-operation and Development (OECD) on CPI inflation rates in India from 1996 to 2022 (31), an effective inflation rate of 6.139 percent is calculated, which is then used to calculate price estimates for 2050. (See Table 2)

Food item	**Price as of 11/3/20 (per kg)**	**Price in 2050 (per kg)**
Brown Rice	34.31	192.1828
Wheat	30.51	29.09
Potato	22.58	21.12
Lentils	83.64	96.726
Milk(UHT/Long-life)	45.49	50.66
Groundnut oil	138.93	181.25

Mustard oil	119.26	189.78
Soybean oil	97.91	161.59
Sunflower oil	105.93	173.24
Jaggery	47.06	48.38
Iodized Salt	15.49	18.95
Onion	30.15	33.52
Tomato	19.2	23.42
Sugar	39.16	41.16

Table 2: Prices of food items in 2020 and 2050 (34)

7.3. **Linear Programming Analysis**

Using price data from Table 2 and nutritional information from Table 1, along with the dietary guidelines from section 1 of this chapter, the objective using 2020 data is to

Minimize

$$\mathbf{0.03431x_1+0.03051x_2+0.02258x_3+0.08364x_4+0.04549x_5+0.13893x_6+0.11926x_7+0.09791x_8+0.10593x_9+0.04706x_{10}+0.01549x_{11}+0.03015x_{12}+0.0192x_{13}+0.03916x_{14}}$$

And using 2050 data is to

Minimize

$$\mathbf{0.192183x_1+0.02909x_2+0.02112x_3+0.096726x_4+0.05066x_5+0.18125x_6+0.18978x_7+0.16159x_8+0.17324x_9+0.04838x_{10}+0.01895x_{11}+0.03352x_{12}+0.02342x_{13}+0.04116x_{14}}$$

With the restrictions being

$$\mathbf{(2.2x_1+1.9x_2+0.1x_3+0.38x_4+3.8x_5+93.4x_6+100x_7+94.6x_8+93.2x_9+0.1x_{10}+0.1x_{12}+o.42x_{13})*9/100<4536000}$$ [fat requirement]

$$\mathbf{x_{11}<35}$$ [salt requirement]

$$\mathbf{7.3x_1+10.6x_2+2x_3+9.02x_4+3.6x_5+0.4x_{10}+0.94x_{12}+0.7x_{13})/100>=336}$$ [protein requirement]

$$\mathbf{x_{14}<=377.0574}$$ [free sugar requirement]

$$\mathbf{(71.1x_1+61.6X_2+15.4x_3+20.1x_4+4.6x_5+9.93x_{12}+3.84x_{13})/100>=91}$$ [carbohydrate requirement]

Where the variables used indicate (in grams), the amounts of Rice(x_1), Wheat(x_2), Potato(x_3), Lentils(x_4), Milk(x_5), Groundnut Oil(x_6), Mustard Oil(x_7), Soybean oil(x_8), Sunflower oil(x_9), Jaggery(x_{10}), Iodized salt(x_{11}), Onion(x_{12}), Tomato(x_{13}), Sugar(x_{14}).

8. References

1) Meadows, D.H. and Meadows, D., 2007. The history and conclusions of The Limits to Growth. System Dynamics Review: The Journal of the System Dynamics Society, 23(2-3), pp.191-197.
2) Vermeulen SJ, Campbell BM, Ingram JSI (2012) Climate change and food systems. Annu Rev Environ Resour 37(1):195–222.
3) Sirohi, S. and Michaelowa, A., 2007. Sufferer and cause: Indian livestock and climate change. Climatic change, 85(3), pp.285-298.
4) Tilman D, Clark M (2014) Global diets link environmental sustainability and human

health. Nature 515(7528):518–522.

5) Hedenus F, Wirsenius S, Johansson DJA (2014) The importance of reduced meat and dairy consumption for meeting stringent climate change targets. Clim Change 124(1-2):79–91.

6) Popp A, Lotze-Campen H, Bodirsky B (2010) Food consumption, diet shifts and associated non-CO2 greenhouse gases from agricultural production. Glob Environ Change 20(3):451–462.

7) Bajlzelj B, et al. (2014) Importance of food-demand management for climate mitigation. Nat Clim Chang 4(10):924–929.

8) Godfray HCJ, et al. (2010) Food security: The challenge of feeding 9 billion people. Science 327(5967):812–818.

9) Hallström E, Carlsson-Kanyama A, Börjesson P (2015) Environmental impact of dietary change: A systematic review. J Clean Prod 91:1–11.

10) Ripple WJ, et al. (2014) Ruminants, climate change and climate policy. Nat Clim Chang 4(1):2–5.

11) Stehfest E, et al. (2009) Climate benefits of changing diet. Clim Change 95(1):83–102.

12) Van Dooren, C., 2018. A review of the use of linear programming to optimize diets, nutritiously, economically and environmentally. *Frontiers in nutrition*, *5*, p.48.

13) Smith, V.E., 1959. Linear programming models for the determination of palatable human diets. *Journal of Farm Economics*, 41(2), pp.272-283.

14) Stigler, G.J., 1945. The cost of subsistence. *Journal of farm economics*, 27(2), pp.303-314.

15) Dantzig, G.B., 1990. The diet problem. *Interfaces*, 20(4), pp.43-47.

16) Buttriss, J.L., Briend, A., Darmon, N., Ferguson, E.L., Maillot, M. and Lluch, A., 2014.

Diet modelling: how it can inform the development of dietary recommendations and public health policy. *Nutrition bulletin*, *39*(1), pp.115-125.

17) Springmann, M., Godfray, H.C.J., Rayner, M. and Scarborough, P., 2016. Analysis and valuation of the health and climate change cobenefits of dietary change. *Proceedings of the National Academy of Sciences*, *113*(15), pp.4146-4151.

18) Wunsch, N.-G., 2021. Number of vegans in Great Britain 2014-2019. [Online] Available at: https://www.statista.com/statistics/1062104/number-of-vegans-in-great-britain/#:~:text=Over%20the%20period%20under%20consideration,to%20600%20thousand%20in%202019.[Accessed 13 March 2022].

19) Ipsos Retail Performance 2020, *Vegan Trends in the U.S*, Ipsos Retail Performance, viewed 21 March 2022, <https://www.ipsos-retailperformance.com/en/vegan-trends/>.

20) Hu, F.B., Otis, B.O. and McCarthy, G., 2019. Can plant-based meat alternatives be part of a healthy and sustainable diet?. *Jama*, *322*(16), pp.1547-1548.

21) Alessandrini, R., Brown, M.K., Pombo-Rodrigues, S., Bhageerutty, S., He, F.J. and MacGregor, G.A., 2021. Nutritional Quality of Plant-Based Meat Products Available in the UK: A Cross-

22) ResearchandMarkets.com, 2021. India Meat Substitutes Market Report 2021: Increasing Vegetarian Population is Expected to Bolster Market Growth to 2027. [Online] Available at: https://www.prnewswire.com/news-releases/india-meat-substitutes-market-report-2021-increasing-vegetarian-population-is-expected-to-bolster-market-growth-to-2027-301255813.html [Accessed 13 March 2022].

23) Sawe, B. E., 2019. Countries With The Highest Rates Of Vegetarianism. [Online] Available at: https://www.worldatlas.com/articles/countries-with-the-highest-rates-of-vegetarianism.html#:~:text=1.,was%20around%206th%20Century%20BC. [Accessed 13 March 2022].

24) National Institute of Nutrition, 2020. NIN's 'My plate for the day' recommends the ideal 2000-calorie diet for every day. [Online] Available at: https://www.nin.res.in/news/paper9.pdf [Accessed 13 March 2022].

25) WHO (2003) Diet, Nutrition and the Prevention of Chronic Diseases: Report of the Joint WHO/FAO Expert Consultation (WHO, Geneva).

26) WHO (2004) Human Energy Requirements: Report of a Joint FAO/WHO/UNU Expert Consultation, Rome, Italy, 17–24 October 2001 (WHO, Geneva).

27) WCRF/AICR (2007) Food, Nutrition, Physical Activity, and the Prevention of Cancer: A Global Perspective (AICR, Washington, DC).

28) Harvard Health Publishing Staff, 2022. How much protein do you need every day?.[Online] Available at: https://www.health.harvard.edu/blog/how-much-protein-do-you-need-every-day-201506188096#:~:text=The%20Recommended%20Dietary%20Allowance%20(RDA,meet%20your%20basic%20nutritional%20requirements. [Accessed 13 March 2022].

29) U.S. Department of Agriculture (USDA), Agricultural Research Service, April 2021. FoodData Central: Foundation Foods. Internet: fdc.nal.usda.gov.

30) Dairy Australia, 2021. Learn more about the different types of milk. [Online] Available at: https://www.dairy.com.au/products/milk [Accessed 13 March 2022].

31) OECD (2022), Inflation (CPI) (indicator). doi: 10.1787/eee82e6e-en (Accessed on 12 April 2022)

32) Lupton, J.R., Brooks, J., Butte, N.F., Caballero, B., Flatt, J.P. and Fried, S.K., 2002. Dietary reference intakes for energy, carbohydrate, fiber, fat, fatty acids, cholesterol, protein, and amino acids. *National Academy Press: Washington, DC, USA*, *5*, pp.589-768.

33) United Nations University and World Health Organization, 2004. *Human Energy Requirements: Report of a Joint FAO/WHO/UNU Expert Consultation: Rome, 17-24*

October 2001 (Vol. 1). Food & Agriculture Org.

34) Price Monitoring Division, 2020. *Daily(Retail and Wholesale Prices)*. [Online] Available at: https://fcainfoweb.nic.in/reports/Report_daily1_Web_New.aspx [Accessed 13 March 2022].